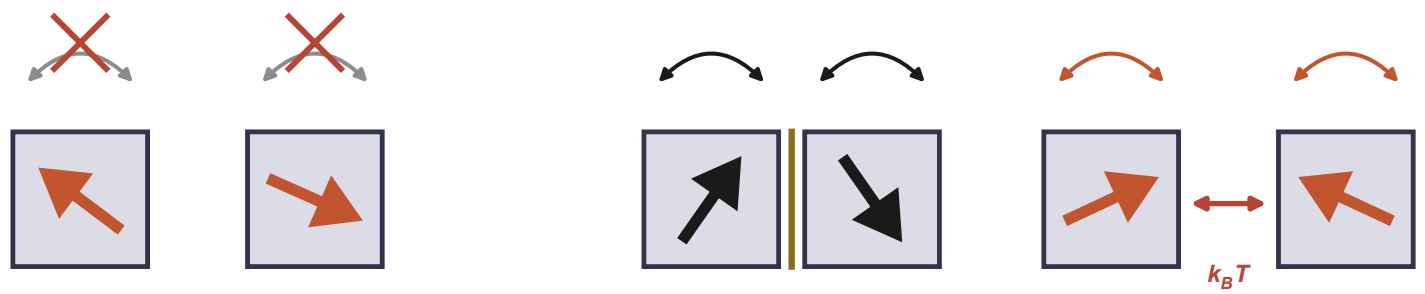


above 268 K



re-sampling, but the well is shallow
→ single phase

Figure 1 is a plot showing the energy components and total energy versus temperature for an ensemble of half-blocked units ($b = 1/2$) at 244 K. The y-axis represents energy in units of $k_B T$, and the x-axis represents temperature in Kelvin.

The plot includes the following components and annotations:

- Legend:**
 - dipolar** $-[(1 - b^2)D + E_{\text{frozen}}]/k_B T$ (Red solid line)
 - van der Waals** $U_{\text{vdW}}/k_B T$ (Blue solid line)
 - aggregation threshold** $-7.24 k_B T$ (Red dashed line)
 - total = van der Waals + dipolar** (Black solid line)
- Annotations:**
 - dipolar WEAKENS on cooling (blocking removes the re-sampling)** (Red arrow pointing to the dipolar curve)
 - van der Waals STRENGTHENS on cooling** (Blue arrow pointing to the van der Waals curve)
- Temperature Range:** A vertical yellow shaded region indicates the temperature range **250-268 K**, where the total energy is below the aggregation threshold.
- Key Temperature:** A vertical dashed line marks $T_B = 250 \text{ K}$ (measured, calibration).
- Ensemble State:** The ensemble is half-blocked ($b = 1/2$) at 244 K.

Figure 1: A plot of normalized energy versus inverse temperature $1/T$. The y-axis is labeled $-E/k_B T$ and ranges from 0 to 1. The x-axis is labeled $1/T$ and ranges from 0 to 1. Three curves are shown: a solid orange line labeled $-(1-b^2)D$ still being re-sampled, a dashed black line labeled $-E_{\text{frozen}}/k_B T$ ceiling, if every moment froze bonded, and a solid black line labeled $-E_{\text{frozen}}/k_B T$ self-consistent. The dashed line starts at approximately 0.5 and increases towards 1. The solid orange line starts at approximately 0.8 and decreases towards 0. The solid black line is a horizontal line at $y=1$. A vertical dashed yellow line is at $1/T \approx 0.5$. A grey shaded region is bounded by the dashed line and the solid black line. A text annotation points to the dashed line: "the self-consistent frozen term never goes below $-0.036 k_B T$ ".

Figure 1 is a line graph showing the temperature dependence of the fraction of particles in the dense phase (x) and the fraction of particles with one or both moments blocked (b and b^2) for a 100% blocked system. The x-axis represents Temperature (K) from 200 to 300. The y-axis is an arbitrary unit. The dashed red line represents b (one moment blocked), the solid red line represents b^2 (both moments blocked), and the blue line represents x (particles in the dense phase). A shaded yellow region highlights the peak in x around 255 K, labeled "peak 13 %".